

Characterization of molecular genetic alterations in GBMs highlights a distinctive molecular profile in young adults.

PMID: 22089350 | Indexed in PubMed

To evaluate age-related differences in histopathologic and molecular profile of glioblastomas (GBMs) at various age groups, with special reference to TP53 mutation, epidermal growth factor receptor (EGFR) amplification, EGFR vIII mutant, PTEN deletion, and IDH1 mutation. Age-wise GBM incidence was calculated over a period of 5 years (2005 to 2009). Seventy-five GBMs were selected for molecular analysis. Majority of cases were in the age group of 41 to 60 years, and mean age was 43.6 years. Histology of all 75 cases selected for molecular profiling was identical. Primary adult GBMs showed EGFR amplification and PTEN deletion in majority (37.3% and 54.9%, respectively). TP53 and IDH1 mutations were rare (11.8% cases each). In secondary GBMs, TP53 (66.7%) and IDH1 mutations (44.4%) were most frequent. PTEN deletion was seen in 33.3% and none had EGFR amplification. Pediatric GBMs (<18 y) harbored frequent TP53 mutations (46.7%) and PTEN deletion in 40%. IDH1 mutations and EGFR amplification were absent. The molecular profile of primary GBMs in young adults (19 to 40 y) was distinctly different from that of adults older than 40 years. TP53 mutation was present in 20% cases. The frequency of EGFR amplification (13.3%) and PTEN deletion (33.3%) was significantly low ($P=0.028$ and 0.046 , respectively). IDH1 mutation, which is rare in primary adult GBMs, was present in 40% of cases. Molecular heterogeneity exists within GBMs of different age cohorts. The molecular profile of GBMs in young adults is distinctly different. Thus, there is a strong need for further studies in various age groups to provide guidelines for therapeutic targeting.

Molecular and Genomic Alterations in Glioblastoma Multiforme.

PMID: 25976245 | Indexed in PubMed

In recent years, important advances have been achieved in the understanding of the molecular biology of glioblastoma multiforme (GBM); thus, complex genetic alterations and genomic profiles, which recurrently involve multiple signaling pathways, have been defined, leading to the first molecular/genetic classification of the disease. In this regard, different genetic alterations and genetic pathways appear to distinguish primary (eg, EGFR amplification) versus secondary (eg, IDH1/2 or TP53 mutation) GBM. Such genetic alterations target distinct combinations of the growth factor receptor-ras signaling pathways, as well as the phosphatidylinositol 3-kinase/phosphatase and tensin homolog/AKT, retinoblastoma/cyclin-dependent kinase (CDK) N2A-p16(INK4A), and TP53/mouse double minute (MDM) 2/MDM4/CDKN2A-p14(ARF) pathways, in cells that present features associated with key stages of normal neurogenesis and (normal) central nervous system cell types. This translates into well-defined genomic profiles that have been recently classified by The Cancer Genome Atlas Consortium into four subtypes: classic, mesenchymal, proneural, and neural GBM. Herein, we review the most relevant genetic alterations of primary versus secondary GBM, the specific signaling pathways involved, and the overall genomic profile of this genetically heterogeneous group of malignant tumors.

Overexpression of the EGF receptor and p53 mutations are mutually exclusive in the evolution of primary and secondary glioblastomas.

PMID: 8864278 | Indexed in PubMed

Glioblastoma multiforme, the most malignant human brain tumor, may develop de novo (primary glioblastoma) or through progression from low-grade or anaplastic astrocytoma (secondary glioblastoma). We present further evidence that primary and secondary glioblastomas constitute distinct disease entities which develop through the acquisition of different genetic alterations. We analyzed p53 mutations, p53 protein accumulation and epidermal growth factor receptor (EGFR) overexpression in 49 biopsies classified as primary or secondary glioblastoma according to clinical and histopathologic criteria. Patients with primary glioblastoma were selected on the basis of a clinical history of less than 3 months and histopathologic features of glioblastoma at the first biopsy (19 cases; mean age, 55 years). The diagnosis of secondary glioblastomas required at least two biopsies and clinical as well as histologic evidence of progression from low grade or anaplastic astrocytoma (30 cases; mean age, 39 years). DNA sequence analysis showed that p53 mutations were rare in primary glioblastomas (11%) while secondary glioblastomas had a high incidence of p53 mutations (67%), of which 90% were already present in the first biopsy. The incidence of p53 protein accumulation (nuclear immunoreactivity to PAb 1801) was also lower in primary (37%) than in secondary glioblastomas (97%). In contrast, immunoreactivity for the EGF receptor prevailed in primary glioblastomas (63%) but was rare in secondary glioblastomas (10%). Only one out of 49 glioblastomas showed EGFR overexpression and a p53 mutation. These data indicate that overexpression of the EGF receptor and mutations of the p53 tumor suppressor gene are mutually exclusive events defining two different genetic pathways in the evolution of glioblastoma as the common phenotypic endpoint.

SEC61G assists EGFR-amplified glioblastoma to evade immune elimination.

PMID: 37523556 | Indexed in PubMed

Amplification of chromosome 7p11 (7p11) is the most common alteration in primary glioblastoma (GBM), resulting in gains of epidermal growth factor receptor (EGFR) copy number in 50 to 60% of GBM tumors. However, treatment strategies targeting EGFR have thus far failed in clinical trials, and the underlying mechanism remains largely unclear. We here demonstrate that EGFR amplification at the 7p11 locus frequently encompasses its neighboring genes and identifies SEC61G as a critical regulator facilitating GBM immune evasion and tumor growth. We found that SEC61G is always coamplified with EGFR and is highly expressed in GBM. As an essential subunit of the SEC61 translocon complex, SEC61G promotes translocation of newly translated immune checkpoint ligands (ICLs, including PD-L1, PVR, and PD-L2) into the endoplasmic reticulum and promotes their glycosylation, stabilization, and membrane presentation. Depletion of SEC61G promotes the infiltration and cytolytic activity of CD8⁺ T cells and thus inhibits GBM occurrence. Further, SEC61G inhibition augments the therapeutic efficiency of EGFR tyrosine kinase inhibitors in mice. Our study demonstrates a critical role of SEC61G in GBM immune evasion, which provides a compelling rationale for combination therapy of EGFR-amplified GBMs.

Mutations of TP53, amplification of EGFR, MDM2 and CDK4, and deletions of CDKN2A in malignant astrocytomas.

PMID: 10323080 | Indexed in PubMed

We investigated the frequency and mutual relationship of molecular alterations in 33 malignant

astrocytomas (28 glioblastomas and 5 anaplastic astrocytomas). The genetic alterations analyzed were: deletion of CDKN2a/p16 gene, TP53 mutations, and amplification of EGFR, MDM2 and CDK4. The most common genetic alteration was EGFR amplification which was revealed in 15 cases (45%). TP53 mutation was identified in 9 cases (27%) and CDKN2/p16 deletion was detected in 13 cases (41%). Either MDM2 and CDK4 amplifications were less frequent, as they were identified in 4 (12%) and 1 (3%) case, respectively. Of the 15 cases showing the amplification of EGFR, 9 had CDKN2/p16 deletion (60%, $p = 0.04$). On the other hand, CDKN2/p16 deletion and EGFR amplification rarely occurred with TP53 mutations (2 of 14 cases with CDKN2/p16 deletion, 14%). These results confirm the existence of at least two different pathways leading to the formation of a glioblastoma.

PIK3CA alterations in primary (de novo) and secondary glioblastomas.

PMID: 17235514 | Indexed in PubMed

We assessed alterations in the EGFR/PTEN/PI3K pathway in 107 primary (de novo) glioblastomas and 32 secondary glioblastomas that progressed from low-grade or anaplastic astrocytomas. SSCP followed by DNA sequencing in exons 9 and 20 of the PIK3CA gene revealed missense mutations in 5/107 (5%) primary and 1/32 (3%) secondary glioblastomas. Quantitative real-time PCR showed PIK3CA amplification (>3 copy numbers) in 14/107 (13%) primary and 3/32 (9%) secondary glioblastomas. Only one glioblastoma showed both PIK3CA mutation and amplification. Taken together with previously published data on EGFR amplification and PTEN mutations, at least one alteration in the EGFR, PTEN, or PIK3CA genes was detected in 63% of primary glioblastomas, which was significantly more frequent than in secondary glioblastomas (31%; $P < 0.001$). Furthermore, this signaling pathway was altered by either PTEN mutations or PIK3CA amplification in 10 of 12 (83%) malignant glioma cell lines analyzed. These results suggest that the EGFR/PTEN/PI3K pathway is frequently altered in glioblastomas and is a promising target for therapy, in particular for primary glioblastomas.

Normal and pathogenic variation of RFC1 repeat expansions: implications for clinical diagnosis.

PMID: 37450567 | Indexed in PubMed

Cerebellar ataxia, neuropathy and vestibular areflexia syndrome (CANVAS) is an autosomal recessive neurodegenerative disease, usually caused by biallelic AAGGG repeat expansions in RFC1. In this study, we leveraged whole genome sequencing data from nearly 10 000 individuals recruited within the Genomics England sequencing project to investigate the normal and pathogenic variation of the RFC1 repeat. We identified three novel repeat motifs, AGGGC ($n = 6$ from five families), AAGGC ($n = 2$ from one family) and AGAGG ($n = 1$), associated with CANVAS in the homozygous or compound heterozygous state with the common pathogenic AAGGG expansion. While AAAAG, AAAGGG and AAGAG expansions appear to be benign, we revealed a pathogenic role for large AAAGG repeat configuration expansions ($n = 5$). Long-read sequencing was used to characterize the entire repeat sequence, and six patients exhibited a pure AGGGC expansion, while the other patients presented complex motifs with AAGGG or AAAGG interruptions. All pathogenic motifs appeared to have arisen from a common haplotype and were predicted to form highly stable G quadruplexes, which have previously been demonstrated to affect gene transcription in other conditions. The assessment of these novel configurations is warranted in CANVAS

patients with negative or inconclusive genetic testing. Particular attention should be paid to carriers of compound AAGGG/AAAGG expansions when the AAAGG motif is very large (>500 repeats) or the AAGGG motif is interrupted. Accurate sizing and full sequencing of the satellite repeat with long-read sequencing is recommended in clinically selected cases to enable accurate molecular diagnosis and counsel patients and their families.

Human African trypanosomiasis in the Democratic Republic of the Congo: disease distribution and risk.

PMID: 26047813 | Indexed in PubMed

For the past three decades, the Democratic Republic of the Congo (DRC) has been the country reporting the highest number of cases of human African trypanosomiasis (HAT). In 2012, DRC continued to bear the heaviest burden of gambiense HAT, accounting for 84 % of all cases reported at the continental level (i.e., 5,968/7,106). This paper reviews the status of sleeping sickness in DRC between 2000 and 2012, with a focus on spatio-temporal patterns. Epidemiological trends at the national and provincial level are presented. The number of HAT cases reported yearly from DRC decreased by 65 % from 2000 to 2012, i.e., from 16,951 to 5,968. At the provincial level a more complex picture emerges. Whilst HAT control in the Equateur province has had a spectacular impact on the number of cases (97 % reduction), the disease has proved more difficult to tackle in other provinces, most notably in Bandundu and Kasai, where, despite substantial progress, HAT remains entrenched. HAT prevalence presents its highest values in the northern part of the Province Orientale, where a number of constraints hinder surveillance and control. Significant coordinated efforts by the National Sleeping Sickness Control Programme and the World Health Organization in data collection, reporting, management and mapping, culminating in the Atlas of HAT, have enabled HAT distribution and risk in DRC to be known with more accuracy than ever before. Over 18,000 locations of epidemiological interest have been geo-referenced (average accuracy ≈ 1.7 km), corresponding to 93.6 % of reported cases (period 2000-2012). The population at risk of contracting sleeping sickness has been calculated for two five-year periods (2003-2007 and 2008-2012), resulting in estimates of 33 and 37 million people respectively. The progressive decrease in HAT cases reported since 2000 in DRC is likely to reflect a real decline in disease incidence. If this result is to be sustained, and if further progress is to be made towards the goal of HAT elimination, the ongoing integration of HAT control and surveillance into the health system is to be closely monitored and evaluated, and active case-finding activities are to be maintained, especially in those areas where the risk of infection remains high and where resurgence could occur.

Review of the Situation of Human African Trypanosomiasis in the Republic of Congo From the 1950s to 2020.

PMID: 35137146 | Indexed in PubMed

Human African trypanosomiasis (HAT), despite considerable progress in the control, is still occurring in many countries in both west and central African regions. The HAT situation in the Republic of Congo has always been overshadowed by its neighbor the Democratic Republic of Congo where over 60% of all HAT cases occur. In the Republic of Congo, HAT cases have been significantly reduced to about 20 reported cases yearly and the disease is still prevalent in few foci across the country. Although continuous

assessment of HAT situation in Congo is been led by the National Control Program for HAT, research on the vector, parasite, and vector control has received little attention. Because there have not been enough reviews summarizing key findings from studies conducted so far, there is still a poor understanding of the global situation of HAT in Congo. In order to achieve sustainable elimination of HAT in Congo a deep appraisal of HAT situation is required. The present study provides a review of studies conducted on HAT in the republic of Congo since the 1950s to date in order to identify gaps in knowledge and help consolidate the gains and progress towards the elimination of sleeping sickness.

Components of event-related potentials and borderline personality disorder: a meta-analysis.

PMID: 38214169 | Indexed in PubMed

Background: Borderline personality disorder (BPD) is characterized by symptoms associated with difficulties in emotion regulation, altered self-image, impulsivity, and instability in personal relationships. A relationship has been found between BPD symptoms and altered neuropsychological processes. Studies of event-related potentials (ERP) measured with electroencephalogram (EEG) have found neural correlates related to BPD symptoms. Of note is the P300 component, considered a potential mental health biomarker for trauma-associated disorders. However, no meta-analysis has been found to demonstrate this relationship. **Objectives:** To evaluate the relationship between the P300 component and BPD symptoms. To evaluate the relationship of other ERP components with BPD symptoms. **Methods:** The method and procedure were adjusted to the PRISMA checklist. The search was performed in three databases: WOS, Scopus and PubMed. A Random Effects Model was used to perform the analysis of the studies. In addition, a meta-regression was performed with % women, Gini and GDP. Finally, a descriptive analysis of the main results found between P300, other ERP components (LPP, P100 and ERN/Ne) and BPD symptoms was performed. **Results:** From a review of 485 articles, a meta-analysis was performed with six articles that met the inclusion criteria. A moderate, positive relationship was found between the P300 component and BPD symptoms ($REM = .489$; $p < .001$). It was not possible to perform meta-analyses for other ERP components (LPP, P100 and ERN/Ne) due to the low number of articles found. **Conclusion:** The idea that P300 could be considered for use as a biomarker to identify altered neural correlates in BPD is reinforced. In addition, a moderating effect of inequality (Gini) was detected.
